# Day 2 Study Notes

Sat 14 Nov 2026 · Sprint 60 · JEE Main 2027

| Part | Time | What to do |
|---|---|---|
| **Study** | 3 h 30 min | Physics: Units, dimensions and errors (75 min) · Maths: 3D geometry (part 1) (75 min) · Chemistry: Coordination compounds (part 1) (60 min) |
| **Exam** | 1 h 15 min | The Day 2 exam in the Sprint 60 app: 30 questions, timed, no notes. |
| **Review** | 1 h 15 min | Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes. |

**How to use each chapter note.** Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

# Physics · Units, dimensions and errors

Usually one to two questions per paper. Questions are short: dimensional formula, error percentage, significant figures.

## 1. SI base units

Seven base quantities: length (m), mass (kg), time (s), electric current (A), temperature (K), amount of substance (mol), luminous intensity (cd). Every other unit is derived from these. In dimensional formulae they are written $M, L, T, A, K$.

## 2. Dimensional formulae to know by heart

| Quantity | Formula used | Dimensions |
|---|---|---|
| Force | $ma$ | $MLT^{-2}$ |
| Work, energy, torque | $Fd$ | $ML^2T^{-2}$ |
| Power | $W/t$ | $ML^2T^{-3}$ |
| Pressure, stress, elastic modulus, energy density | $F/A$ | $ML^{-1}T^{-2}$ |
| Momentum, impulse | $mv$ | $MLT^{-1}$ |
| Angular momentum, Planck's constant $h$ | $mvr$, $E/\nu$ | $ML^2T^{-1}$ |
| Gravitational constant $G$ | $Fr^2/m^2$ | $M^{-1}L^3T^{-2}$ |
| Surface tension, spring constant | $F/l$ | $MT^{-2}$ |
| Coefficient of viscosity | $F/(A\,dv/dx)$ | $ML^{-1}T^{-1}$ |
| Boltzmann constant | $E/T$ | $ML^2T^{-2}K^{-1}$ |
| Charge | $It$ | $AT$ |
| Potential difference | $W/q$ | $ML^2T^{-3}A^{-1}$ |
| Resistance | $V/I$ | $ML^2T^{-3}A^{-2}$ |
| Capacitance | $q/V$ | $M^{-1}L^{-2}T^4A^2$ |
| Inductance | $V/(dI/dt)$ | $ML^2T^{-2}A^{-2}$ |
| Magnetic field $B$ | $F/(qv)$ | $MT^{-2}A^{-1}$ |
| Permittivity $\varepsilon_0$ | $q^2/(Fr^2)$ | $M^{-1}L^{-3}T^4A^2$ |
| Permeability $\mu_0$ | $2\pi rF/(I^2l)$, force between wires | $MLT^{-2}A^{-2}$ |

**Combinations the exam likes.** $\dfrac{1}{\sqrt{\mu_0\varepsilon_0}}$ and $\dfrac{E}{B}$ are speeds. $RC$, $\dfrac{L}{R}$ and $\sqrt{LC}$ are times. $\dfrac{h}{e^2}$ is a resistance. $\frac{1}{2}\varepsilon_0 E^2$ and $\dfrac{B^2}{2\mu_0}$ are energy densities, the same dimensions as pressure. Angles, strain, refractive index and relative density have no dimensions.

## 3. Three uses of dimensions

1. **Checking an equation.** Every term added or subtracted must have the same dimensions. The argument of $\sin$, $\cos$, $\log$ and $e^{x}$ must be dimensionless.
2. **Deriving a relation.** Assume $Q = k\,a^{x}b^{y}c^{z}$, equate powers of $M$, $L$, $T$ on both sides, and solve. The constant $k$ cannot be found this way.
3. **Converting units.** $n_1u_1 = n_2u_2$, so $n_2 = n_1\left(\dfrac{M_1}{M_2}\right)^{a}\left(\dfrac{L_1}{L_2}\right)^{b}\left(\dfrac{T_1}{T_2}\right)^{c}$.

> **Solved example 1.** The time period of a liquid drop depends on density $\rho$, radius $r$ and surface tension $S$. Find the relation.
> Let $T = k\rho^{x}r^{y}S^{z}$. Dimensions: $T = (ML^{-3})^{x}L^{y}(MT^{-2})^{z}$. Comparing powers: $M$: $x + z = 0$; $L$: $-3x + y = 0$; $T$: $-2z = 1$. So $z = -\tfrac{1}{2}$, $x = \tfrac{1}{2}$, $y = \tfrac{3}{2}$, giving $T = k\sqrt{\dfrac{\rho r^{3}}{S}}$.

## 4. Significant figures

- All non-zero digits count. Zeros between non-zero digits count.
- Leading zeros never count: $0.00420$ has three significant figures.
- Trailing zeros after a decimal point count: $4.200$ has four.
- Multiplying or dividing: keep as many significant figures as the least precise number.
- Adding or subtracting: keep as many decimal places as the number with the fewest.

## 5. Errors

- Mean value $a_m = \dfrac{\sum a_i}{n}$. Absolute error of each reading $\Delta a_i = |a_i - a_m|$.
- Mean absolute error $\Delta a = \dfrac{\sum \Delta a_i}{n}$. Relative error $= \dfrac{\Delta a}{a_m}$. Percentage error $= \dfrac{\Delta a}{a_m} \times 100$.
- Sum or difference $Z = A \pm B$: absolute errors add, $\Delta Z = \Delta A + \Delta B$.
- Product or quotient: relative errors add, $\dfrac{\Delta Z}{Z} = \dfrac{\Delta A}{A} + \dfrac{\Delta B}{B}$.
- Powers, $Z = \dfrac{A^{p}B^{q}}{C^{r}}$: $\dfrac{\Delta Z}{Z} = p\dfrac{\Delta A}{A} + q\dfrac{\Delta B}{B} + r\dfrac{\Delta C}{C}$.

> **Solved example 2.** $g = \dfrac{4\pi^{2}L}{T^{2}}$. If $L$ is known to 1% and $T$ to 2%, the error in $g$ is $1\% + 2 \times 2\% = 5\%$. The power of $T$ doubles its contribution, and errors always add even when the quantity is in the denominator.

**Traps.** Errors never subtract. For $A - B$ the absolute errors still add, which makes the percentage error large when $A$ and $B$ are close. The quantity with the highest power dominates the error, so it must be measured most carefully. Torque and work have the same dimensions but are different quantities. A dimensionally correct equation is not necessarily physically correct.

**Read in NCERT:** Physics Class 11 Part 1, chapter "Units and Measurements" (significant figures, dimensions and dimensional analysis sections, with the solved examples). Tomorrow covers measuring instruments: vernier calipers, screw gauge and least count.

# Mathematics · 3D geometry (part 1)

3D geometry was one of the most-asked Maths chapters of 2026, at almost two questions per paper. The syllabus covers lines only, not planes. Today: points, direction cosines and the equation of a line. Tomorrow: skew lines, shortest distance, foot of perpendicular and image.

## 1. Points in space

- Distance between $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$: $PQ = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$.
- Point dividing $PQ$ internally in ratio $m : n$: $\left(\dfrac{mx_2 + nx_1}{m+n}, \dfrac{my_2 + ny_1}{m+n}, \dfrac{mz_2 + nz_1}{m+n}\right)$. For external division replace $n$ by $-n$.
- Quick method for "in what ratio does a coordinate plane divide $PQ$": take the ratio as $k : 1$ and set the relevant coordinate to zero. For the $yz$-plane, $x = 0$ gives $k = -\dfrac{x_1}{x_2}$.
- Distance of $(x, y, z)$ from the $x$-axis is $\sqrt{y^2 + z^2}$; from the $xy$-plane it is $|z|$.

## 2. Direction cosines and direction ratios

- If a line makes angles $\alpha, \beta, \gamma$ with the positive axes, its direction cosines are $l = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$, and $l^2 + m^2 + n^2 = 1$.
- It follows that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$ and $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$.
- Direction ratios $a, b, c$ are any three numbers proportional to $l, m, n$. Then $l = \dfrac{a}{\sqrt{a^2+b^2+c^2}}$, and similarly for $m$, $n$.
- Direction ratios of the line through $P$ and $Q$: $x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1$.
- A line equally inclined to the three axes has direction cosines $\pm\dfrac{1}{\sqrt{3}}, \pm\dfrac{1}{\sqrt{3}}, \pm\dfrac{1}{\sqrt{3}}$.

## 3. Angle between two lines

For direction ratios $(a_1, b_1, c_1)$ and $(a_2, b_2, c_2)$:

$$\cos\theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2+b_1^2+c_1^2}\,\sqrt{a_2^2+b_2^2+c_2^2}}$$

- Perpendicular lines: $a_1a_2 + b_1b_2 + c_1c_2 = 0$.
- Parallel lines: $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$.
- Three points $A$, $B$, $C$ are collinear when the direction ratios of $AB$ and $AC$ are proportional.

## 4. Equation of a line

| Given | Vector form | Cartesian form |
|---|---|---|
| Point $\vec{a} = (x_1, y_1, z_1)$ and direction $\vec{b} = (a, b, c)$ | $\vec{r} = \vec{a} + \lambda\vec{b}$ | $\dfrac{x-x_1}{a} = \dfrac{y-y_1}{b} = \dfrac{z-z_1}{c}$ |
| Two points $\vec{a}$ and $\vec{b}$ | $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$ | $\dfrac{x-x_1}{x_2-x_1} = \dfrac{y-y_1}{y_2-y_1} = \dfrac{z-z_1}{z_2-z_1}$ |

**The one technique that solves most line questions.** Write a general point on the line as $(x_1 + a\lambda,\ y_1 + b\lambda,\ z_1 + c\lambda)$, apply the condition in the question to find $\lambda$, and substitute back. This handles where a line meets a plane, intersection of two lines, and tomorrow's foot of perpendicular.

**Solved example 1.** Find where the line through $A(3, 4, 1)$ and $B(5, 1, 6)$ crosses the $xy$-plane.
Direction ratios: $(2, -3, 5)$. General point: $(3 + 2\lambda,\ 4 - 3\lambda,\ 1 + 5\lambda)$. On the $xy$-plane $z = 0$, so $\lambda = -\frac{1}{5}$. The point is $\left(\frac{13}{5}, \frac{23}{5}, 0\right)$.

**Solved example 2.** Find the angle between the lines $\dfrac{x}{2} = \dfrac{y}{2} = \dfrac{z}{1}$ and $\dfrac{x-5}{4} = \dfrac{y-2}{1} = \dfrac{z-3}{8}$.
$\cos\theta = \dfrac{|8 + 2 + 8|}{\sqrt{9}\sqrt{81}} = \dfrac{18}{27} = \dfrac{2}{3}$, so $\theta = \cos^{-1}\dfrac{2}{3}$.

## 5. Projection

The projection of the segment $PQ$ on a line with direction cosines $l, m, n$ is $|(x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n|$. Convert direction ratios to direction cosines first.

**Traps.** Before reading direction ratios from a Cartesian equation, make the coefficients of $x$, $y$, $z$ equal to $+1$. For example $\dfrac{1 - x}{3}$ must be rewritten as $\dfrac{x - 1}{-3}$, and $\dfrac{2x - 1}{4}$ as $\dfrac{x - \frac{1}{2}}{2}$. Direction ratios are not unique but direction cosines are fixed up to sign. Two angles with the axes fix the third only up to $\gamma$ or $180^\circ - \gamma$.

**Read in NCERT:** Mathematics Class 12 Part 2, chapter "Three Dimensional Geometry": direction cosines and direction ratios, equation of a line in space, angle between two lines, with the solved examples.

# Chemistry · Coordination compounds (part 1)

The single most-asked Chemistry chapter of 2026, at about two questions per paper. Today: Werner's theory, ligands and naming. Day 3: isomerism. Day 4: bonding, magnetism and colour.

## 1. The vocabulary

| Term | Meaning | In $[Co(NH_3)_5Cl]Cl_2$ |
|---|---|---|
| Central atom | Metal that accepts electron pairs | Co |
| Ligands | Ions or molecules that donate electron pairs to the metal | five $NH_3$, one $Cl^-$ |
| Coordination sphere | Metal plus ligands, written in square brackets; does not ionise | $[Co(NH_3)_5Cl]^{2+}$ |
| Counter ions | Ions outside the bracket; ionise in water | two $Cl^-$ |
| Coordination number | Number of donor atoms bonded to the metal | 6 |
| Oxidation state | Charge on the metal if all ligands are removed with their electron pairs | +3 |

A complex with one kind of ligand is **homoleptic**, for example $[Co(NH_3)_6]^{3+}$. With more than one kind it is **heteroleptic**, for example $[Co(NH_3)_4Cl_2]^+$.

## 2. Werner's theory

- **Primary valence** is ionisable, is satisfied by negative ions, and equals the oxidation state.
- **Secondary valence** is non-ionisable, is satisfied by ligands, is fixed for a metal, equals the coordination number, and is directed in space (this gives the complex its shape).

| Compound | Werner formula | AgCl precipitated per mole | Ions in solution |
|---|---|---|---|
| $CoCl_3 \cdot 6NH_3$ | $[Co(NH_3)_6]Cl_3$ | 3 mol | 4 |
| $CoCl_3 \cdot 5NH_3$ | $[Co(NH_3)_5Cl]Cl_2$ | 2 mol | 3 |
| $CoCl_3 \cdot 4NH_3$ | $[Co(NH_3)_4Cl_2]Cl$ | 1 mol | 2 |

Only chloride outside the bracket is precipitated by $AgNO_3$. More ions in solution means higher molar conductivity.

**Double salt or complex?** A double salt such as Mohr's salt, $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$, breaks completely into simple ions in water. A complex such as $K_4[Fe(CN)_6]$ keeps its complex ion intact, so the solution gives no test for $Fe^{2+}$ or $CN^-$.

## 3. Types of ligand

| Type | Donor atoms | Examples |
|---|---|---|
| Monodentate | 1 | $Cl^-$, $OH^-$, $CN^-$, $H_2O$, $NH_3$, CO |
| Didentate | 2 | ethane-1,2-diamine (en), oxalate $C_2O_4^{2-}$ (ox) |
| Polydentate | more than 2 | $EDTA^{4-}$ is hexadentate (2 N and 4 O) |

| | | |
|---|---|---|
| Ambidentate | 1 at a time, from a choice of 2 | $NO_2^-$ (through N or O), $SCN^-$ (through S or N) |

A di- or polydentate ligand that binds through two or more atoms forms a ring called a **chelate**. Chelate complexes are more stable than similar complexes of monodentate ligands.

**Coordination number counts donor atoms, not ligands.** In $[Fe(C_2O_4)_3]^{3-}$ there are three ligands but the coordination number is 6. In $[Co(en)_3]^{3+}$ it is 6. In $[Fe(EDTA)]^-$ one ligand gives coordination number 6.

**Finding the oxidation state.** Sum of charges equals the charge on the complex. For $K_3[Fe(C_2O_4)_3]$: the complex ion is 3−, so $x + 3(-2) = -3$ and $x = +3$. For $[Ni(CO)_4]$: CO is neutral, so Ni is 0.

## 4. IUPAC naming

1. Name the cation first, then the anion, whichever is the complex.
2. Inside the complex, name ligands in alphabetical order, then the metal. Ignore di, tri, tetra when ordering.
3. Anionic ligands end in **-o**: chlorido, cyanido, hydroxido, oxalato, nitrito-N (through N), nitrito-O (through O), thiocyanato-S or thiocyanato-N.
4. Neutral ligands keep their names, with four exceptions: aqua ($H_2O$), ammine ($NH_3$), carbonyl (CO), nitrosyl (NO).
5. Use di, tri, tetra for simple ligands. Use bis, tris, tetrakis when the ligand name already contains a number prefix, with the ligand in brackets: tris(ethane-1,2-diamine).
6. Give the oxidation state of the metal in Roman numerals in brackets.
7. If the complex ion is an anion, the metal name ends in **-ate**: cobaltate, chromate, nickelate, platinate, and from Latin names ferrate (Fe), cuprate (Cu), argentate (Ag), aurate (Au).

| Formula | Name |
|---|---|
| $[Co(NH_3)_5Cl]Cl_2$ | pentaamminechloridocobalt(III) chloride |
| $K_4[Fe(CN)_6]$ | potassium hexacyanidoferrate(II) |
| $[Ni(CO)_4]$ | tetracarbonylnickel(0) |
| $[Pt(NH_3)_2Cl(NO_2)]$ | diamminechloridonitrito-N-platinum(II) |
| $[Co(en)_3]_2(SO_4)_3$ | tris(ethane-1,2-diamine)cobalt(III) sulphate |
| $K_3[Al(C_2O_4)_3]$ | potassium trioxalatoaluminate(III) |

**Traps.** "Ammine" for $NH_3$ has two m's; "amine" is an organic group. The number of counter ions is never stated in the name: work it out from the charges. In formulas, the metal is written first, then ligands. A neutral complex is named as one word with no -ate ending.

**Read in NCERT:** Chemistry Class 12 Part 1, chapter "Coordination Compounds": Werner's theory, definitions of important terms, and nomenclature. Read every worked example and attempt the in-text questions.